



Assignment 11

By Wilner Harel

1

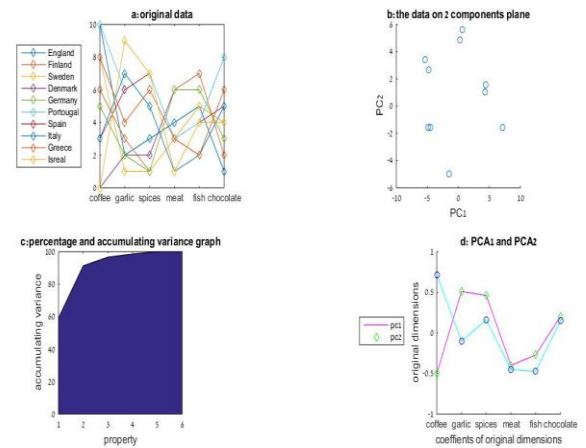


Figure 1: a- food consumption of different countries, b- PCA and plot of data in 2-dimension, c- percentage and accumulating variance graph, d- values of PC1 versus PC2

1A:

It seems that the countries :England& Sweden, Portugal& Greece, Italy& Spain and Finland& Germany are all clustered in couples .

2

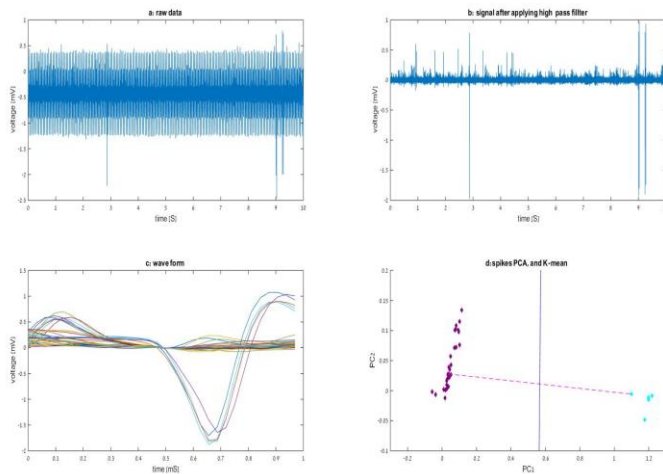


Figure 2: a- original data, b- b. afret filtering the noise (frequency below 300), c- wave form, d- spikes wave form and K-means clustering, separation line(blue) is perpendicular bisector to the line between to closet points (pink)

2E

According to the graphs, the noise correlation between neurons 14 and 17 varies between the 8 directions.

While directions 1-3 have a positive correlation, the others' correlations are negative. (the highest correlation is in the third direction which is close to 1).

1C

In Pc1 it seems that coffee as the highest value while fish as the lowest, on Pc2 it seems that Garlic as the highest value while coffee as the lowest. The PC are built from combination of the original basis, in the first Pc there is more weight to the property with the highest variance and in the second pc there is more weight to the property with second highest variance.

The original variables all correlated to some degree with one another (positively or negatively), the pc are linear transformation between them that as zero correlation with each other, so we can describe the data with linear transformation of original data according to the degree of variance of each variable

Woo-Hoo!